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Kelas : 1A TNK

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1. Diberikan persamaan : $y = 3\sin x + 3\sqrt{3}\cos x$, dan ubahlah dalam bentuk $y = R\sin(x \pm \alpha)$

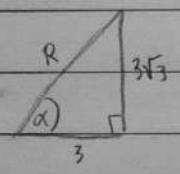
$3\sin x + 3\sqrt{3}\cos x = R\sin(x+\alpha)$

$3\sin x + 3\sqrt{3}\cos x = R[\sin x \cos\alpha + \cos x \sin\alpha]$

$3\sin x + 3\sqrt{3}\cos x = R\sin x\cos\alpha + R\cos x \sin\alpha)$

$3\sin x = R\sin x\cos\alpha \rightarrow \cos\alpha = \frac{3}{R}$

$3\sqrt{3}\cos x = R\cos x \sin\alpha \rightarrow \sin\alpha = \frac{3\sqrt{3}}{R}$



$R = \sqrt{a^2+b^2} = \sqrt{3^2+(3\sqrt{3})^2} = \sqrt{9+27} = \sqrt{36} = 6$

$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = \frac{3\sqrt{3}/R}{3/R} = \frac{3\sqrt{3}}{3} = \sqrt{3}$

$\alpha = \arctan\sqrt{3}$

$\alpha = 60°$

Jadi, $3\sin x + 3\sqrt{3}\cos x = 6\sin(x+60°)$

a. Berapa periode dan amplitudo y

periode 2π, amplitudo $y = 6$

b. Berapa y pada $x = 60°$

$y = 6\sin(x+60°)$

$y = 6\sin(60°+60°)$

$y = 6\sin 120°$

$y = 6 \cdot 0{,}866$

$y = 5{,}196$

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c. Tentukan dua buah x, jika $y = 5$

$6 \sin (x + 60^\circ) = 5$

$\sin (x + 60^\circ) = \frac{5}{6}$

$x + 60^\circ = \text{arc} \sin \frac{5}{6}$

$x + 60^\circ = 56,44^\circ$

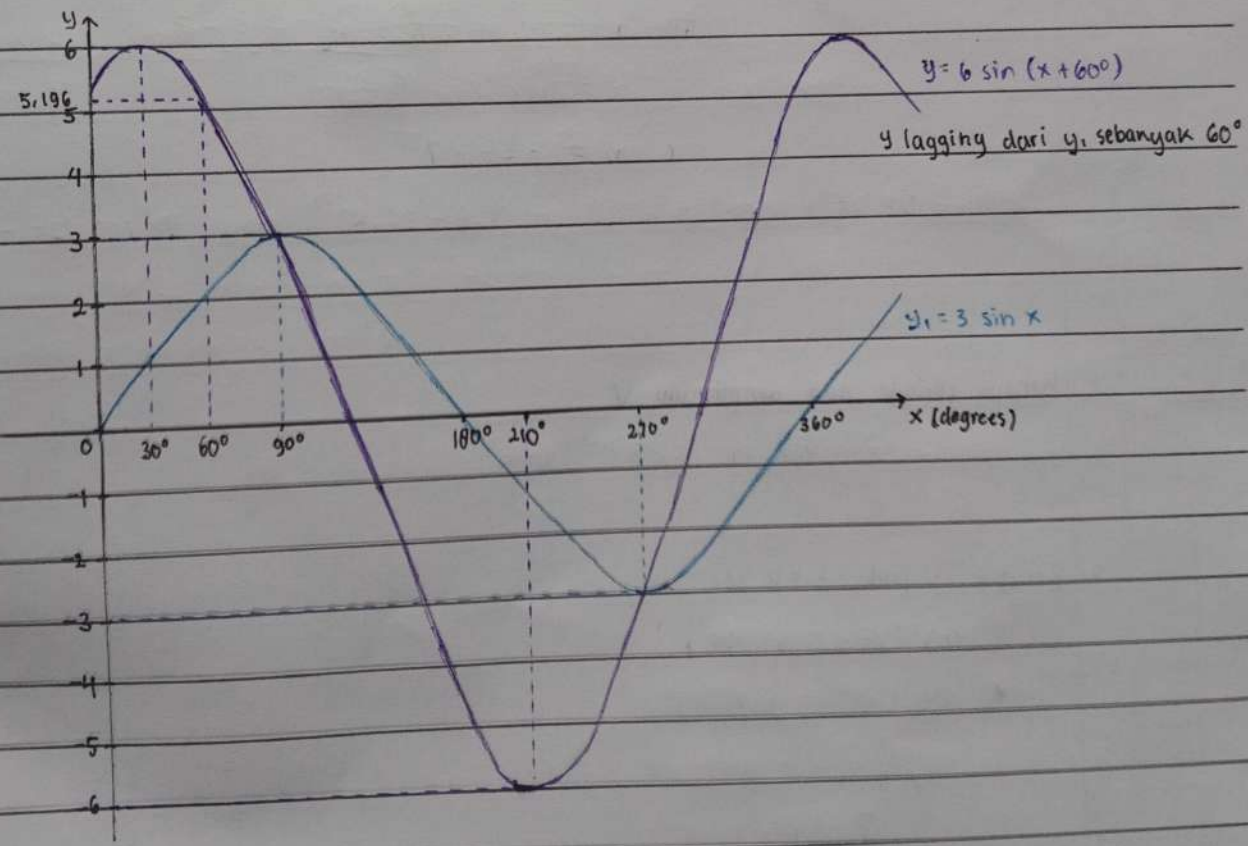
$x = 56,44^\circ - 60^\circ$

$x_1 = -3,56^\circ$

$x_1 = -3,56^\circ$

$x_2 = -3,56^\circ + 360^\circ = 356,44^\circ$

d. Sketsa grafik y dengan acuan $y_1 = 3 \sin x$ (y lagging atau leading y_1)



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2. Diketahui persamaan ; $V_1 = 6 \sin \omega t$ dan $V_2 = 8 \cos \omega t$. Jika tegangan memenuhi persamaan $V = V_1 + V_2$, dengan $\omega = 2$ rad / sec, t dalam sec.

a. Ubahlah / nyatakan V dalam bentuk $R \cos (2t \pm \alpha)$

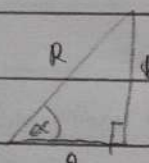
$6 \sin 2t + 8 \cos 2t = R \cos (2t + \alpha)$

$6 \sin 2t + 8 \cos 2t = R (\cos 2t \cos \alpha - \sin 2t \sin \alpha)$

$6 \sin 2t + 8 \cos 2t = R \cos 2t \cos \alpha - R \sin 2t \sin \alpha)$

$6 \sin 2t = -R \sin 2t \sin \alpha \rightarrow \sin \alpha = -\frac{6}{R}$

$8 \cos 2t = R \cos 2t \cos \alpha \rightarrow \cos \alpha = \frac{8}{R}$



$R = \sqrt{a^2 + b^2} = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$

$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{-6/R}{8/R} = -\frac{6}{8} = -\frac{3}{4}$

$\alpha = \arctan (-\frac{3}{4})$

$\alpha = -36,87° \rightarrow$ KW IV $= 360° - 36,87° = 323,13°$

Jadi, $6 \sin 2t + 8 \cos 2t = 10 \cos (2t - 323,13°)$ atau $10 \cos (2t - 5,63968$ radian)

b. Berapa periode dan amplitudo V

periode $\Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$ detik, amplitudo = 10

c. Berapa V pada $t = 4$ sec

$V = 10 \cos (2t - 5,63968)$

$= 10 \cos (2(4) - 5,63968)$

$= 10 \cos (8 - 5,63968)$

$= 10 \cos 2,36032$

$= 10 \times (-0,71001)$

$= -7,1001$ volt

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d. Kapan t bila V = 7 volt

$10 \cos (2t - 5{,}63968) = 7$

$\cos (2t - 5{,}63968) = \frac{7}{10}$

$2t - 5{,}63968 = \text{arc cos } \frac{7}{10}$

$2t - 5{,}63968 = 0{,}79539$

$2t = 0{,}79539 + 5{,}63968$

$2t = 6{,}43507$

$t = \frac{6{,}43507}{2} = 3{,}2117535$

e. Sketsa grafik V dengan acuan V_2 (lagging atau leading V_2)

